LAB 7
TECH 60 - Sec 14
F26
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Objective:
The objective of this lab is to understand the relationship between voltages and currents in a series/parallel circuit and to confirm Ohm's Law, Kirchhoff's Voltage Law (KVL), Kirchhoff's Current Law (KCL), the Voltage Divider Rule, and the Current Divider Rule. We will do this by comparing hand calculations, a Multisim simulation, and breadboard measurements.

Equipment/Materials:
Solderless Breadboard
Keysight EDU36311A, Triple Output, Programmable DC Power Supply
Keysight EDU34450A 5½ Digit Digital Multimeter
Multisim software running on PC
220 Ω resistors (Quantity 1)
330 Ω resistors (Quantity 1)
470 Ω resistors (Quantity 1)
1 kΩ resistors (Quantity 1)
Jumper wires

Part I. Hand Calculations
Vs = 9 V
R_1 = 330 Ω
R_2 = 1 kΩ = 1000 Ω
R_3 = 470 Ω
R_4 = 220 Ω

Total resistance
R_2‖3 = (R_2 × R_3)/(R_2 + R_3)
(1000Ω × 470Ω)/(1000Ω + 470Ω) = 470,000Ω/1470 = 319.73 Ω
R_T = R_1 + R_2‖3 + R_4
330Ω + 319.73Ω + 220Ω = 869.73 Ω

Total current (Ohm's Law)
I_T = V_s/R_T
9V / 869.73 Ω = 10.348 mA
I_1 = I_4 = I_T

Voltage drops (Ohm's Law / Voltage Divider Rule)
V_1 = I_T × R_1 = *0.010348A × 330Ω = 3.415 V*
V_4 = I_T × R_4 = *0.010348A × 220Ω = 2.277 V*

$V_2 = V_3 = V_{2\|3} = I_T \times R_{2\|3} = 0.010348A \times 319.73\Omega = 3.309\ V$

Branch currents (Current Divider Rule)

$I_2 = I_T \times R_3/(R_2 + R_3) = 0.010348A \times 470/1470 = 3.309\ mA$

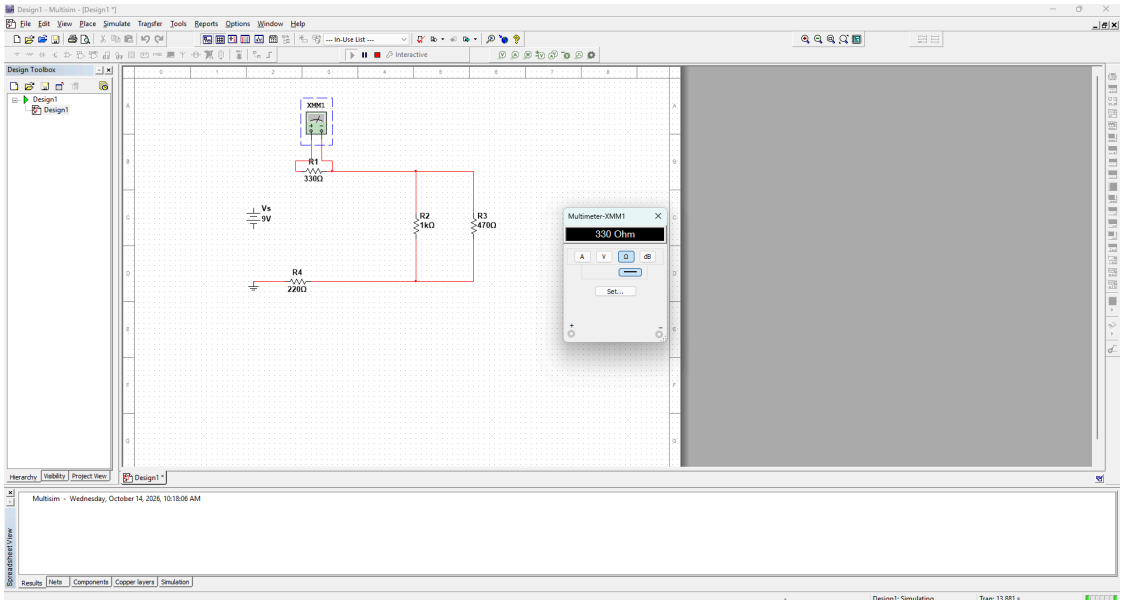
$I_3 = I_T \times R_2/(R_2 + R_3) = 0.010348A \times 1000/1470 = 7.040\ mA$

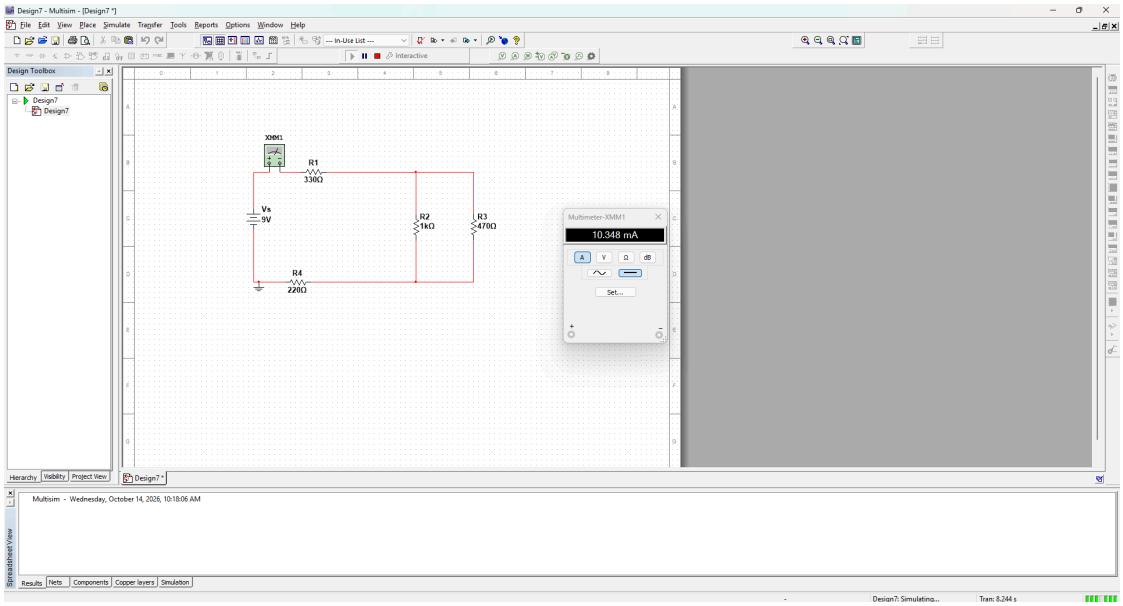
Check (KCL and KVL)

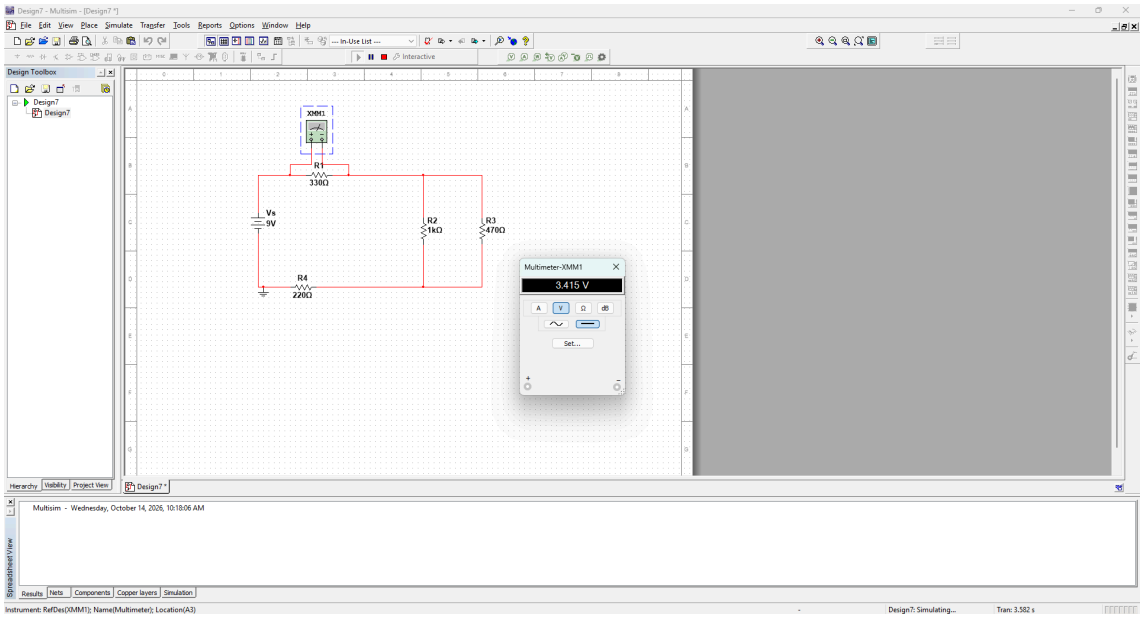
$I_1 = I_2 + I_3$: $10.348\ mA = 3.309\ mA + 7.039\ mA$

$Vs = V_1 + V_{2\|3} + V_4$: $9\ V = 3.415\ V + 3.309\ V + 2.277\ V = 9.001\ V \approx 9\ V$

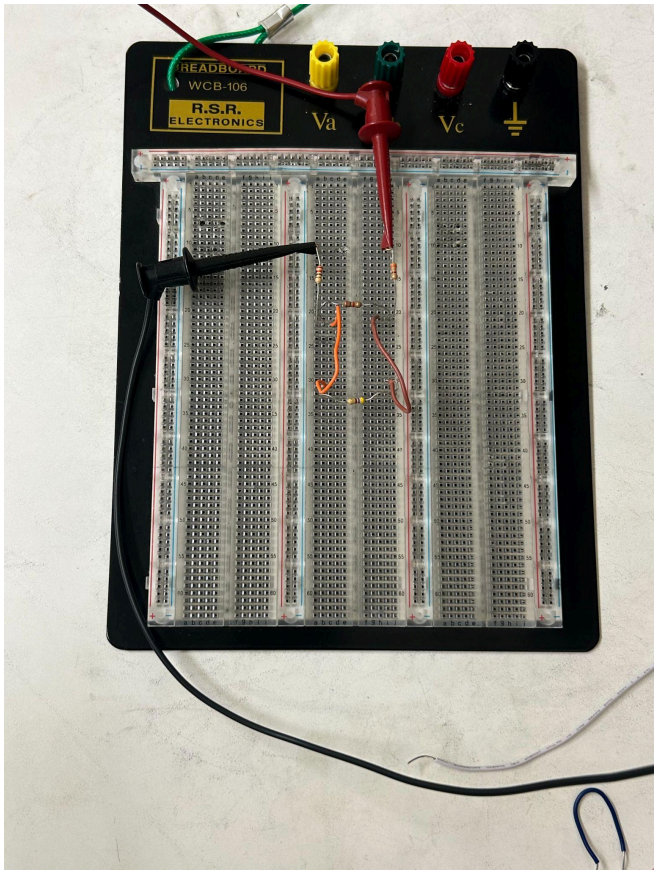
Part II. Circuit simulation using Multisim

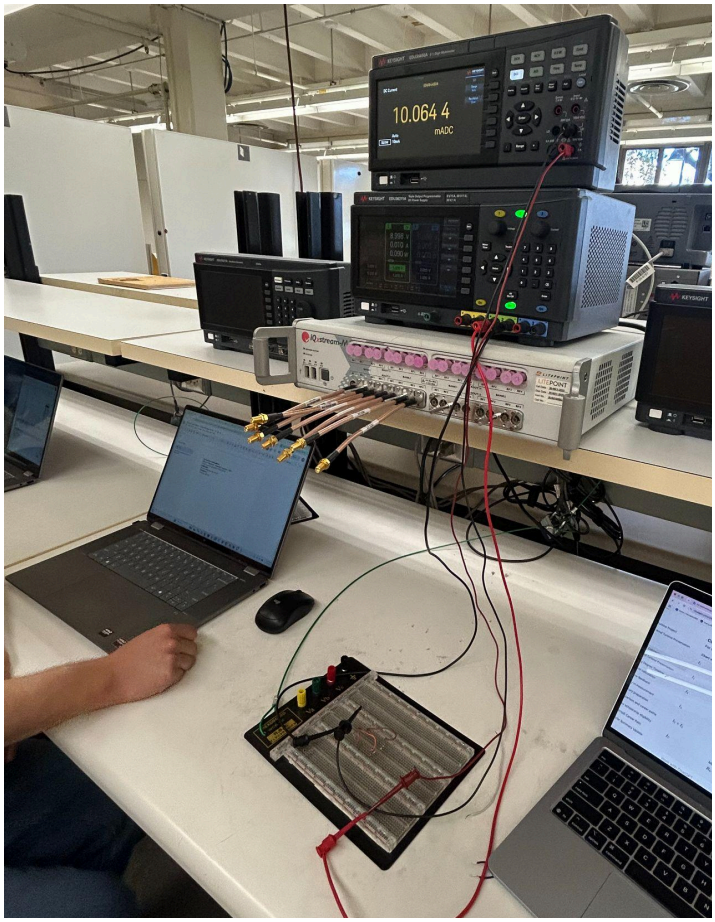






Part III. Circuit construction on the breadboard







Part IV. Questions

1. Compare the measured resistor values to the expected values. If they are not exactly the same then explain briefly why they are not exactly equal.

The measured resistor values were slightly different from the expected values. For example, R1 was expected to be 330 Ω but measured 328 Ω. These differences can be caused by resistor tolerances, temperature changes, and small measurement errors.

2. What do you observe regarding the relationship between the current I1, and the current I4?

I1 and I4 theoretically are the same value, in the breadboard data they come within 1.8% of each other. The two currents are also equal to the total current I_T.

3. What do you observe regarding the relationship between the current I1, and the sum of currents I2 and I3?

The current I1 is equal to the sum of I2 and I3. This follows Kirchhoff's Current Law (KCL), which states that the total current entering a junction must equal the total current leaving it.

4. What do you observe regarding the relationship between the voltage V2 and V3?

The voltages V2 and V3 are equal because both resistors are connected in parallel. According to our calculations, both voltages are 3.309 V.

5. How is the voltage VS divided across resistors of the circuit?

The 9 V source voltage is divided among the series portions of the circuit. R1 has 3.415 V, the parallel branch containing R2 and R3 has 3.309 V, and R4 has 2.277 V. These voltage drops add up to approximately 9 V.

6. Explain how Kirchhoff's Voltage Law (KVL) is satisfied in this circuit.

KVL is satisfied since V1 + V2||3 + V4 in all the different data sections (hand calculations, multisim, and breadboard) add up to V_s.

7. Explain how Kirchhoff's Current Law (KCL) is satisfied in this circuit.

KCL is confirmed through the currents in each branch, I1 is 10.064, and it splits off into I2 and I3 through the first node. I2 and I3 add up to 10.37 which is within 3% of the initial current in branch 1.

8. Write a paragraph about the relationship between the current flows in the circuit and how it divides in the parallel branch.

The same current flows through the series segments of the circuit. The total current leaves the supply and passed through R1, and R4 so I1=I4=IT. Then the

current splits between R2 and R3 inversely proportional to the resistance values of R2 and R3. The branch currents join again at the bottom node.

9. Compare the measured total current to the calculated, and the Multisim value of this quantity. Should they be the same value ideally? Explain briefly why they are not exactly equal.

The calculated and Multisim total currents were both 10.348 mA, and the measured breadboard value was 10.25 mA, which means we had a 0.95% error. Ideally, we should have gotten the same values but due to human error, breadboard resistance, and jumper wire resistance, we can expect some percent error.

10. From your experimental results have you gained a good understanding of series/parallel circuit?

Yes, the results showed that in a series portion, the same current flows through R1 and R4, and that in the parallel section R2 and R3 have the same voltage while the current divided between them inversely proportional to their resistance values. The breadboard experiment showed this as well with some understandable percent error due to the factors listed above in the previous question.

Figure 1. Resistance Data

	R1	R2	R3	R4	RT
Expected value	330 Ω	1 kΩ	470 Ω	220 Ω	869.7 Ω
Measured	328 Ω	.987 kΩ	467 Ω	218 Ω	862 Ω
%Error Between Expected Value and Measured	0.6%	1.30%	0.64%	0.91%	0.89%

Figure 2. Current Data

	I1	I2	I3	I4	I2 + I3	IT
Hand Calculations	10.348 mA	3.309 mA	7.039 mA	10.348 mA	10.348 mA	10.348 mA
Multisim	10.348 mA	3.309 mA	7.039 mA	10.348 mA	10.348 mA	10.348 mA
Breadboard	10.064 mA	3.38 mA	6.99 mA	10.25 mA	10.37 mA	10.25 mA

	I1	I2	I3	I4	I2 + I3	IT
%Error Between Hand Calculations and Breadboard	2.74%	2.15%	0.70%	0.95%	0.21%	0.95%

Figure 3. Voltage Data

	V1	V2	V3	V4	V1 + V2‖3 + V4	VS
Hand Calculations	3.415 V	3.309 V	3.309 V	2.277 V	9.001 V	9 V
Multisim	3.415 V	3.309 V	3.309 V	2.277 V	9.001 V	9 V
Breadboard	3.315 V	3.34 V	3.12 V	2.24 V	8.895 V	8.998 V
%Error Between Hand Calculations and Breadboard	2.93%	0.94%	5.71%	1.62%	1.18%	0.02%

Percent Error Calculations

%Error = |Measured − Expected| / Expected × 100%

Challenges:

One challenge we faced during this lab was connecting the resistors correctly on the breadboard, especially when working with both series and parallel connections. We couldn't find a way to connect the resistors correctly and we couldn't get the numbers that we wanted. Another challenge was using the digital multimeter to measure voltage and current accurately. We also had difficulty understanding why our measured values were slightly different from our hand calculations and Multisim results. These differences could have been caused by resistor tolerances, loose connections, or measurement errors. Lastly, applying Ohm's Law and Kirchhoff's Laws to calculate the total resistance, current, and voltage required careful calculations to avoid mistakes.

Conclusion:

In conclusion, this lab helped us better understand how voltage and current behave in series and parallel circuits. We used hand calculations, Multisim simulations, and a breadboard circuit to compare our results. Our hand calculations and Multisim results showed a total resistance of approximately 869.73 Ω and a total current of 10.348 mA. However, our breadboard measurements showed a total resistance of 862 Ω and a

current of 10.064 mA. These small differences may be caused by resistor tolerances, connection issues, or measurement errors. We also learned how Ohm's Law, Kirchhoff's Voltage Law, and Kirchhoff's Current Law apply to series and parallel circuits. Overall, this lab helped us improve our understanding of circuit behavior and gain experience using electronic equipment.